

AYRA ERP System Requirements Specification

Project: AYRA ERP Portal

1. Introduction

AYRA ERP is a web-based academic enterprise resource planning system for students, teachers, academic office staff, and communication teams. It manages student records, subjects, classes, advisories, leave requests, approvals, calendar events, and email alerts.

2. Main Modules

- Student: profile, progress, classes, advisories, calendar, leave requests, photo upload
- Teacher: student management, subjects, classes, advisories, progress, leave approval, alerts
- Academic Office: teachers, curriculum, timetables, approvals, records, leave monitor, alerts
- Communication: announcements, campaigns, events, response tracking

3. Alert System

- Email alerts use Nodemailer with Gmail app password configuration
- Alerts can be sent to all students, department, semester, section, or specific student emails
- Alert history is stored in MongoDB with recipient count and delivery status

4. Technology Stack

- Frontend: React 18, Vite 5, React Router DOM 6, Material UI 5, Tailwind CSS 3
- Backend: Node.js, Express 4, Mongoose 8, Nodemailer 6
- Database: MongoDB Community / MongoDB Atlas

5. Functional Requirements

- Role-based login and tenant-aware routes
- CRUD operations for academic and ERP data
- Student email storage for targeted communication
- Academic calendar and timetable management
- Leave request and approval workflow
- Communication and alert broadcast support

6. Non-Functional Requirements

- Responsive user interface for desktop and mobile
- Database-backed persistence and validation
- Environment-based configuration for mail and MongoDB
- Modular frontend and backend architecture

7. Theme, Font, and Color Information

- Primary Font Family: Manrope, Segoe UI, sans-serif
- MUI Typography Font Family: "Manrope", "Segoe UI", sans-serif
- h1 weight: 800, letter spacing: -0.04em
- h2 weight: 800, letter spacing: -0.03em
- h3 weight: 700
- Button font weight: 700, textTransform: none

8. Core Theme Colors

- Primary Main: #0f766e
- Primary Light: #ccfbf1
- Primary Dark: #115e59
- Secondary Main: #1d4ed8
- Background Default: #f4f7f4
- Background Paper: #ffffff
- Text Primary: #102a26
- Text Secondary: #47635d

9. UI Styling Details